포스터발표7

RGB 영상과 Mask R-CNN을 이용한 생육 초기 잡초 자동 동정

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Automatic Early-stage Weed Identification Using Field RGB Images and Mask R-CNN

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Weeds compete with crops for essential resources such as water, light, and nutrients, inhibiting crop growth and ultimately reducing both yield and quality. Therefore, managing these weeds is crucial, and effective control in the early stages of growth is particularly important as it can prevent weeds from preempting vital resources. However, weeds at this stage are difficult to identify visually due to their morphological similarities. In this context, this study was conducted to establish a system that automatically recognizes and distinguishes weed plants at the early stage of growth by using Mask R-CNN. RGB images were collected from maize and soybean fields in 2021, and the dataset was composed of 5 weed species found in the fields such as *Chenopodium album, Digitaria ciliaris, Portulaca oleracea, Abutilon theophrasti*, and *Echinochloa crus-galli*. From the various RGB images collected, five weeds and two crops were extracted as individual plants using SAM (Segment Anything Model). Augmentation techniques were applied to enlarge the training dataset. After training the model with mask R-CNN algorithm, the model performance was evaluated through Precision, Recall, F_1 score, and IoU (Intersection of Union) based on the testset. Our results demonstrated that the model effectively predicted the type and region of weeds in the field. This study indicates that the deep learning-based weed identification system can be applied even in the early stages of weed growth, and suggests its potential for broader application to various weed species and growth stages.

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